

Research Proposal

MSc Thesis in Data Science and Business Analytics

Connecting the Islands: Automated Integration of Ferry and Road Networks for
Multimodal Accessibility Analysis

by

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Introduction

The accessibility of transportation in islands and coastal areas not only depends on the quality of road infrastructure, but also on the effective connection between water transport and the road transportation system. In many areas, ferries are an essential means of transportation. If without ferries, islands could be isolated from the mainland's resources, such as healthcare, education, employment, and public services. In digital transportation modeling, ferry and road networks are often derived from heterogeneous spatial data sources with different geometric precision, data structures, and topological properties. Therefore, they cannot always be directly integrated into a coherent multimodal transport network. This can lead to misaligned terminal locations, missing transfer nodes, and disconnected graph components. This can also distort shortest path results and bias in accessibility assessments. As a result, before accessibility can be measured reliably, the underlying ferry and road networks must first be integrated in a robust and automated way (Chen, 2005).

This thesis covers the interdisciplinary fields of geospatial data integration, transportation network modeling, and accessibility analysis. From a geospatial perspective, the primary challenge lies in data conflation. Most of conflation studies focus on road-to-road matching or on aligning roads with imagery, rather than on integrating road and ferry systems into a unified network for accessibility analysis (Chen, 2005; Zhang et al., 2016). The reliability of accessibility analysis depends on the transportation network it is based on. If islands are left disconnected in the network model, or if ferry and road transfer points are poorly represented, then the resulting accessibility estimates may understate or distort actual reachability. In the other side, the ferry routes are not automatically included in standard road-network extraction workflows and must be added separately before islands can be reconnected to the mainland network (Krishnakumari, den Hertog and Gromicho, n.d.)

This paper aims to study how ferry and road networks from heterogeneous spatial data sources can be automatically integrated into a unified multimodal transport network. The central idea is that accessibility analysis in island and coastal regions does not begin with shortest-path calculations, but with a robust spatial integration procedure that identifies and resolves misalignment between transport layers.

Research questions

How can ferry and road networks from heterogeneous spatial data sources be automatically integrated into a unified multimodal transport network for accessibility analysis?

Methodology and Techniques

1. Spatial data integration / map conflation
2. Nearest-neighbour matching
3. KD-tree / spatial indexing
4. Topological validation
5. Shortest path algorithms

Data and/or expected results

Open and semi-open transport data sources such as OpenStreetMap contain rich multimodal information, including roads and ferry routes. This makes it feasible to build large-scale transportation networks. Data from Nordic countries, the Philippines, or Indonesia will be used in this paper.

The expected result of this paper is to develop an automated integration method for heterogeneous ferry-road data. This model should improve the reliability of accessibility analysis in areas where ferries play a structural role in transportation connections.

Literature

Chen, C.-C., Knoblock, C.A. and Shahabi, C. (2007) 'Automatically and Accurately Conflating Raster Maps with Orthoimagery,' *GeoInformatica*, 12(3), pp. 377–410. <https://doi.org/10.1007/s10707-007-0033-0>.

Krishnakumari, P.K., den Hertog, D. and Gromicho, J. (n.d.) Hands-on Geospatial Analytics for the Social Good: Volume One: Data Journeys. Unpublished manuscript.

Xu, Z. and Chopra, S.S. (2023) 'Interconnectedness enhances network resilience of multimodal public transportation systems for Safe-to-Fail urban mobility,' *Nature Communications*, 14(1), p. 4291. <https://doi.org/10.1038/s41467-023-39999-w>.

Zhang, M., Yao, W. and Meng, L. (2016) 'Automatic and Accurate Conflation of Different Road-Network Vector Data towards Multi-Modal Navigation,' *ISPRS International Journal of Geo-Information*, 5(5), p. 68. <https://doi.org/10.3390/ijgi5050068>.

Hackeloer, A. (2016) Conflation of Road Networks from Digital Maps. <https://www.gbv.de/dms/tib-ub-hannover/884497070.pdf>.

Smarzaro, R., Davis, C. A., & Quintanilha, J. A. (2021). Creation of a Multimodal Urban Transportation Network through Spatial Data Integration from Authoritative and Crowdsourced Data. *ISPRS International Journal of Geo-Information*, 10(7), 470. <https://doi.org/10.3390/ijgi10070470>

Timetable

1st Month

Activity	Date
Discuss research question and data	1. 08-04-2026
Discuss Literature Review and dive into data	2. 22-04-2026

2nd Month

Activity	Date
Apply feedbacks and discuss model	3. 06-05-2026
Discuss the methodology	4. 20-05-2026

3rd Month

Activity	Date
Modify model and apply feedback	5. 03-06-2026
Share and discuss results	6. 17-06-2026

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Master's Thesis Business Analytics



Supervisor

Signature

Joaquim Gromicho

Second marker (suggestion by supervisor)

Dick den Hertog

Thesis coordinator

C.G.H. Diks

Signature